

Artificial Intelligence

Artificial Intelligence (AI) is a branch of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence. These tasks include learning, reasoning, problem-solving, perception, and language understanding.

Machine Learning

Machine Learning is a subset of Artificial Intelligence that enables computers to learn from data without being explicitly programmed. Machine learning algorithms improve their performance as they are exposed to more data.

Deep Learning

Deep Learning is a specialized branch of Machine Learning that uses artificial neural networks with multiple layers. It is widely used in image recognition, speech recognition, natural language processing, and autonomous vehicles.

Natural Language Processing

Natural Language Processing (NLP) enables computers to understand, interpret, and generate human language. Large Language Models (LLMs) such as GPT, Gemini, and Llama are examples of NLP applications.

Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) combines information retrieval with Large Language Models. Instead of relying only on the model's training data, RAG retrieves relevant information from external documents and uses that information to generate more accurate responses.

Applications of AI

Artificial Intelligence is used in healthcare, finance, education, manufacturing, cybersecurity, agriculture, transportation, and entertainment. AI assists doctors in diagnosis, detects fraud in banking, recommends products in e-commerce, and powers virtual assistants.

Future of AI

The future of Artificial Intelligence includes intelligent agents, autonomous robots, multimodal AI systems, personalized education, and advanced scientific research. Responsible AI development focuses on fairness, transparency, privacy, and security.